VO NGUYEN DANG KHOA

SECURITY CONSULTANT & SYSTEMS ARCHITECT

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STRIDE / GRC
RUST / GO

Information Security • Threat Modeling & DRM • Cryptographic Controls • Systems Engineering

GRID SYSTEM // ISO-216 A4

/01 THREAT POSTURE

Risk & Controls

Systems and security architect with 9+ years in technology and 4+ years engineering defense-in-depth controls directly into production software: Ed25519 offline cryptographic licensing, Argon2id + RFC 6238 TOTP authentication, forensic anti-leak DRM, canonicalized filesystem sandboxing, and dual-architecture static Musl toolchains. Combines low-level systems engineering (Rust, Go, Linux kernel subsystems) with regional digital governance across 6 Southeast Asian markets (Procter & Gamble) to align threat models, compensating controls, and supply-chain assurance.

7+ Apps	Ed25519	Zero-Leak	<10MB
PRODUCTION SECURITY PACKAGES	CRYPTOGRAPHIC CONTROLS	FORENSIC ANTI-LEAK DRM	MEMORY-SAFE RUST DAEMONS
Synology DSM 7.2+ Ecosystem	Argon2id + TOTP 2FA Sandboxing	Dynamic Watermarking & Blanking	Rootless /dev/uinput Subsystem

01. Threat Modeling & Non-Normative Flows
Built forensic anti-leak DRM (ola) and kernel input isolation (vietc), designing runtime countermeasures against screen recording, DevTools inspection, and input race conditions.

02. Defense-in-Depth & Cryptography
Engineered Argon2id password hashing, RFC 6238 TOTP 2FA, Ed25519-signed offline licensing, and dunce::canonicalize filesystem sandboxing in kv-file-pro.

03. Compensating Controls & Supply Chain
Solved legacy GLIBC platform limitations via dual-arch static Musl cross-compilation in kv-trimui; enforced automated GnuPG package signing and dual-mirror CI/CD release assurance.

/02 CLOUD PLATFORM

syno.vndns.vn // SPK

kv-synology — Flagship DSM Web Control Plane

[Next.js 15 · React 19 · QuickConnect · GnuPG CI/CD]

https://syno.vndns.vn

Automated SPK repository with GnuPG cryptographic signature verification, dynamic port allocation (20000–59999), QuickConnect relay resolver, and live daemon control (SMB, NFS, SSH, WebDAV, AFP).

ola — Forensic Anti-Leak DRM
Dynamic moving watermark overlay (IP, timestamp, ID) and auto-blank traps upon screen-recording or DevTools inspection.
Go • React • Synology SPK

kv-file-pro — Cryptographic Sandboxing
Argon2id hashing, RFC 6238 TOTP 2FA, Ed25519 asymmetric offline licensing, and dunce::canonicalize path isolation.
Rust (Axum) • SQLite WAL

SysVis.AI — Threat Model Visualizer
Architecture topology and threat boundary mapping for Architecture Review Forums with local-first WebGPU AI inference.
React 19 • WebLLM Qwen3

kv-server — Zero-Trust Tunnel Gateway
Handheld reverse-proxy enforcing Cloudflare Zero-Trust Tunnels, token-based auth (?auth=token), and loopback port isolation.
Rust • Musl • Cloudflared

kv-tube — Hardened Media Engine
Per-IP rate limiting (300 req/min), strict CORS origin allow-lists, and multi-service container network segmentation.
Go (Gin) • Next.js • HLS

spkrepo — Supply Chain Packaging
Automated SPK package distribution platform with GnuPG cryptographic signature verification and container validation.
GnuPG • Docker • Python

/03 KERNEL SUBSYSTEM

VietC // Rust 1.85+

VietC (Viet+) — Native High-Performance Linux Input Method Engine

github.com/vndangkhoa/vietc (60★)

Direct Wayland Virtual Keyboard
Direct Unicode keystroke injection via wtype (zwp_virtual_keyboard_v1) and /dev/uinput. Eliminates pre-edit underline flashing and preserves IDE autocompletion in VS Code and GPU terminals.
wtype • /dev/uinput • <1ms Latency

Intelligent Auto-Restore
Dual-layer Vietnamese phonological analysis paired with a technical English lexicon automatically preserves programming keywords (search, test, filter) without manual mode toggling.
Phonology Engine • Context Awareness

Hardware Event Filtering
Strict device path binding to /dev/input/by-path/*-event-kbd eliminates duplicate keystrokes caused by multi-interface 2.4GHz wireless USB dongles. Consumes <10MB RAM as a rootless daemon.
Hardware Binding • <10MB Footprint

/04 MILESTONE LOG

2020 — Present

2024 — PRES — Secure Architecture & Forensic DRM (Ola & KV Files PRO)
+ Engineered forensic anti-leak DRM (ola) with dynamic moving watermarks (IP/time/ID) and auto-blank traps upon screen-recording or DevTools inspection.
+ Built KV Files PRO with Argon2id password hashing, RFC 6238 TOTP 2FA, offline Ed25519 asymmetric licensing, and dunce::canonicalize filesystem sandboxing.

2024 — PRES — Memory Safety & Kernel Defense (VietC & KV-TrimUI)
+ Architected pure Rust 1.85+ input engine (vietc) with direct kernel /dev/uinput and /dev/input/by-path isolation, eliminating keystroke sniffing and duplicate injection (<10MB rootless).
+ Engineered KV-TrimUI dual-arch static Musl toolchains (aarch64 & armv7l) to overcome legacy GLIBC 2.28 vulnerabilities across 6 handheld OS distributions.

2020 — 2025 — Regional Digital Asset Governance & Compliance (Procter & Gamble SEA)
+ Governed brand risk policies, asset validation, and regulatory compliance across 6 Southeast Asian regional markets (P&G Hair Care portfolio).
+ Architected enterprise modular design system of master components, accelerating adaptation 3× while eliminating non-compliant asset deviations.

/05 TECHNICAL SPEC

Matrix & Endpoints

SECURITY & RISK
Threat Modeling · STRIDE · Non-Normative Flows · Forensic DRM · Watermarking · GRC · Policy Enforcement

CRYPTOGRAPHY & CONTROLS
Argon2id · RFC 6238 TOTP · Ed25519 · GnuPG · Sandboxing · Rate Limiting · CORS · Zero-Trust

SYSTEMS & LOW-LEVEL
Rust 1.85+ · Linux (Arch/Debian) · /dev/uinput · Static Musl · POSIX C · Bash · systemd rootless

CLOUD & STORAGE
Synology DSM 7.2+ · Docker Multi-Arch · Go (Gin) · React 19 · Next.js 15 · Btrfs RAID · SQLite WAL

Storage & Snapshot Architecture
Btrfs RAID storage pools with immutable snapshot replication, encrypted volume allocation, and automated retention lifecycle policies.

Gateway & Routing Infrastructure
Nginx reverse proxy with automated wildcard Let's Encrypt SSL certificate rotation, Hairpin-NAT, and Cloudflare DDNS routing.

Verified Endpoints
https://syno.vndns.vn · github.com/vndangkhoa/vietc · github.com/vndangkhoa/kv-synology

VO NGUYEN DANG KHOA

EMBEDDED SYSTEMS & HANDHELD OS ARCHITECT

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DUAL-ARCH MUSL
/DEV/FB0 60FPS
6 OS DISTROS
Ho Chi Minh City, VN

Linux Framebuffer (/dev/fb0) • Dual-Arch Static Musl • 6 Handheld OS Distros • MPV & ALSA Hi-Fi

GRID SYSTEM // ISO-216 A4

/01 SYSTEM THESIS

A523 & A133P OS

Creator and lead systems architect of KV-TrimUI (trimui.vndns.net)—the complete open-source native software ecosystem and 18-application suite for TrimUI handheld consoles (Smart Pro S TG5050 Allwinner A523 Octa-Core, Smart Pro TG5040 A133P, Brick Pro/Brick). Engineered a Dual-Engine Architecture combining a 100% pure Rust 60 FPS direct framebuffer engine (/dev/fb0 mmap, <900KB–5MB RAM) with lightweight Next.js 15 + dual-arch static Musl daemons (aarch64 & armv7l, <15MB RAM), turnkey packagers across all 6 major OS distributions (Stock OS, SpruceOS, Knulli, NextUI, NXRedux, GammaOS), and a 1-Click Web Installer & dual-mirror OTA distribution network.

18 Apps	6 OS Distros	< 900 KB	Dual-Arch
PRODUCTION HANDHELD SUITE	**MULTI-OS TURNKEY SUPPORT**	**PURE RUST 60 FPS ENGINE**	**STATIC MUSL TOOLCHAINS**
Tube, Tidal, Netflix, TV, Radio, Play, Chat	Stock, Spruce, Knulli, NextUI, NXRedux, Gamma	Direct /dev/fb0 mmap • Zero X11/Wayland	aarch64 & armv7l • A523 & A133P SoCs

/02 KERNEL PILLARS

Dual-Engine · 6-OS

SUBSYSTEM 01

Dual-Engine Architecture (Pure Rust FB + Web Hybrid)

Engine B renders 100% native Rust directly to /dev/fb0 via mmap at 60 FPS (<900KB–5MB RAM) with DRM Plane 93 fail-safe; Engine A pairs Next.js 15 static UIs with static Musl daemons (<15MB RAM) and @kv-trimui/shared-gamepad OSK.

SUBSYSTEM 02

Dual-Arch Static Musl & 6-OS Turnkey Packagers

Cross-compiles zero-dependency static binaries for both aarch64-unknown-linux-musl and armv7-unknown-linux-musleabihf to bypass GLIBC 2.28 limits across Stock OS, SpruceOS, Knulli, NextUI, NXRedux, and GammaOS.

SUBSYSTEM 03

Hardware MPV 720p60 & Bit-Perfect ALSA Hi-Fi

Streams YouTube, cinema, and IPTV via GPU-accelerated MPV (--vo=gpu, Cedar/Mali) and routes Tidal Hi-Fi, Apple Music & AirPlay directly to ALSA hw:0,0 with A523 DACL unity gain (160) tuning.

/03 HANDHELD SUITE

18 Production Apps

kv-radio [PURE RUST 60 FPS FRAMEBUFFER · <900KB RAM]

Rust 1.85 · /dev/fb0 mmap · Direct ALSA

Standalone embedded internet radio station player directly rendering to the Linux framebuffer with double-buffered page flipping, custom TrueType glyph rasterization, dirty-rectangle blitting, and direct ALSA PCM streaming.

kv-tube & kv-tv — YOUTUBE & LIVE IPTV
Ad-free InnerTube YouTube client + 24/7 Live IPTV (VTV, HTV, NHK, Bloomberg) with 720p60 MPV GPU playback.
Next.js 15 • Rust FB • MPV HWDEC

kv-tidal & kv-airplay — HI-FI & AIRPLAY AUDIO
Bit-perfect lossless streamer (Tidal, Apple Music, NAS Vault) + Shairport Sync wireless AirPlay receiver via ALSA hw:0,0.
Rust Musl • Direct ALSA • DACL 160

kv-netflix & kv-stremio — HANDHELD CINEMA
On-demand movie & anime streaming clients with 6-digit phone pairing, episode selectors, and bilingual EN/VI subtitles.
Native Rust /dev/fb0 • HLS • MPV

kv-play & kv-theme — CLOUD ROMS & STYLER
Ozone-styled cloud ROM store with auto .7z/.cue extraction & self-healing cores + real-time handheld theme/icon switcher.
100% Native Rust /dev/fb0 • Multi-OS

kv-news & kv-anime — READER & MANGA 60FPS
Kindle-style ad-free news reader (VnExpress, Tuổi Trẻ) and NetTruyen manga reader with aspect scaling & D-Pad paging.
100% Native Rust Framebuffer

kv-chat, server & torch — AI, TUNNEL & UTILITY
Voice AI companion (Safari WebGPU/LLMs), Cloudflare Tunnel web server daemon, and QR phone-paired LED marquee strobe.
Rust • Cloudflared • WebGPU • WS

/04 FIRMWARE LOG

2026 Roadmap

2025 — PRESENT
KV-TrimUI 18-App Ecosystem & Multi-OS Packaging Pipeline

System Architect & Creator

+ Architected 18 production apps (kv-tube, kv-tidal, kv-netflix, kv-radio, kv-tv, kv-news, kv-anime, kv-stremio, kv-play, kv-theme, kv-server, kv-airplay, kv-chat, kv-torch) for Smart Pro S (A523), Smart Pro (A133P), and Brick.
+ Built automated multi-OS packaging (pnpm package:stock/spruce/knulli/nextui/nxredux & cook_gamma.sh) plus cook.sh single-source dual-mirror OTA release workflow (GitHub + MyDS Forgejo).

2025 — PRESENT
Native Rust Framebuffer Engine, Shared Gamepad & AI Knowledge Hub

Core Engine & Localization Lead

+ Engineered <900KB pure Rust /dev/fb0 engine (kv-radio), compile-time bilingual EN/VI localization (kv-i18n & shared-i18n), and @kv-trimui/shared-gamepad with instant X-button On-Screen Keyboard.
+ Published machine-readable AI Agent Knowledge Hub (llms-full.txt, /api/v1/agent/knowledge, and trimui-rust-starter template) documenting DRM Plane 93 fail-safes and Evdev polling.

/05 EMBEDDED SPEC

Matrix & OTA Links

HARDWARE & SOC TARGETS
Allwinner A523 Octa-Core · Allwinner A133P · Smart Pro S (TG5050) · Brick Pro / Brick · DRM Plane 93 Fix · PWM Fan & Hall Sticks

DUAL-ENGINE & COMPILERS
Rust 1.85+ (/dev/fb0 mmap) · aarch64 & armv7l Musl · Next.js 15 Static · @kv-trimui/shared-gamepad · kv-i18n (EN/VI) · TrueType Rasterizer

MEDIA & KERNEL SUBSYSTEMS
MPV GPU (--hwdec=auto) · ALSA Direct PCM (hw:0,0) · A523 DACL Unity Gain 160 · Linux Evdev Polling · Shairport AirPlay · Safari WebGPU LLM

6-OS PACKAGING & OTA
Stock OS · SpruceOS · Knulli (Batocera) · NextUI / MinUI (.pakz) · NXRedux OS · GammaOS (Android 13) · cook.sh Dual-Mirror OTA

https://trimui.vndns.net
Official App Store & 1-Click Web Installer (/installer)

github.com/vndangkhoa/kv-trimui
Primary Release & Dual-Mirror MyDS Forgejo Repo

trimui.vndns.net/llms-full.txt
Developer & AI Agent Framebuffer Knowledge Hub